

DEFINITION

BECAUSE n IS DISCRETE ONLY IN TIME, NOT IN FREQUENCY
 THE DISCRETE TIME FOURIER TRANSFORM OF A DISCRETE TIME SEQUENCE $x[n]$ IS A REPRESENTATION OF THE SEQUENCE IN TERMS OF COMPLEX EXPONENTIAL SIGNALS.

BEING A TRANSFORM, IT DOES NOT ADD OR REMOVE INFORMATION ABOUT THE SEQUENCE.

EXPRESSES THE SIGNAL AS A FUNCTION OF SINES AND COSINES

$$X(e^{j\omega}) := \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$$

MAPS A DISCRETE SEQUENCE TO A CONTINUOUS ONE

CONCEPTUALLY, THE DTFT ALLOWS US TO CHECK HOW MUCH OF A TONAL COMPONENT AT FREQUENCY ω IS PRESENT IN $x[n]$.

THE DTFT IS DEFINED AS THE INNER PRODUCT BETWEEN THE SIGNAL AND THE COMPLEX EXPONENTIAL TONE $e^{-j\omega n}$.

SINCE COMPLEX EXPONENTIALS $\{e^{-j\omega n}, \omega \in [-\pi, \pi]\}$ FORM A COMPLETE FAMILY OF ORTHOGONAL BASES FOR PRACTICALLY EVERY SIGNAL OF INTEREST, THE DTFT ESSENTIALLY PROJECTS $x[n]$ ONTO THE SPACE SPANNED BY ALL COMPLEX EXPONENTIALS.

THE FUNCTION IS COMPLEX, THEREFORE

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\angle X(e^{j\omega})} = \operatorname{Re}\{X(e^{j\omega})\} + j \operatorname{Im}\{X(e^{j\omega})\}$$

INVERSE

THE DTFT OF A SEQUENCE, IF IT EXISTS, IS UNIQUE AND THEREFORE THE ORIGINAL SEQUENCE CAN BE RETRIEVED USING THE INVERSE TRANSFORM.

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$X(e^{j\omega})$ IS CONTINUOUS!

PROPERTIES

PERIODICITY

THE TRANSFORMED SIGNAL IS ALWAYS PERIODIC OF 2π

$$X(e^{j(\omega+2k\pi)}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j(\omega+2k\pi)n} = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} e^{-j2k\pi n} = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = X(e^{j\omega})$$

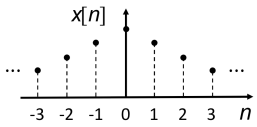
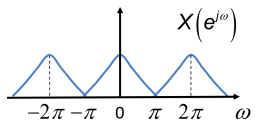
$k \in \mathbb{Z}$

$$e^{j\varphi} = \cos(\varphi) + j \sin(\varphi) \Rightarrow e^{-j2k\pi n} = \cos(2k\pi n) - j \sin(2k\pi n) = 1 - j0 \neq x_{k,n}$$

THUS ALL THE INFORMATION OF THE SIGNAL LIES IN $[0, 2\pi]$ OR $[-\pi, \pi]$

CONTINUITY

THE DTFT MAPS A DISCRETE AND (GENERALLY) APERIODIC SEQUENCE INTO A CONTINUOUS AND PERIODIC SIGNAL

Discrete Time Signals	
Time Domain	Frequency Domain
 $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$	 $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$
Discrete and [generally] aperiodic	Continuous and periodic

SPOILER: THE DFT IS A SAMPLED SEQUENCE OF THE DFT, SO THAT IT IS DISCRETE TOO

LINEARITY

THE DFT IS A LINEAR OPERATION, THEREFORE THE DFT OF A SUM OF SIGNALS IS THE SUM OF THE DFTS OF THE SINGLE SIGNALS.
IT ALSO OBEYS SCALING.

$$\left. \begin{array}{l} x_1[n] \rightarrow X_1(e^{j\omega}) \\ x_2[n] \rightarrow X_2(e^{j\omega}) \end{array} \right\} Ax_1[n] + Bx_2[n] \rightarrow AX_1(e^{j\omega}) + BX_2(e^{j\omega})$$

COMPLEX SEQUENCES

IN CASE OF COMPLEX SEQUENCES:

EVEN SEQUENCES ARE ALSO CALLED "CONJUGATE SYMMETRIC"

ODD SEQUENCES ARE ALSO CALLED "CONJUGATE ANTI-SYMMETRIC"

$$x_{\text{EVEN}}[n] = \frac{1}{2} (x[n] + x^*[-n])$$

SO THAT ALSO IMAGINARY PART IS SYMMETRIC

$$x_{\text{ODD}}[n] = \frac{1}{2} (x[n] - x^*[-n])$$

$$x[n] = x_{\text{EVEN}}[n] + x_{\text{ODD}}[n]$$

$$x_{\text{EVEN}}[n] = x_{\text{EVEN}}^*[-n]$$

$$x_{\text{ODD}}[n] = -x_{\text{ODD}}^*[-n]$$

$$X_{\text{EVEN}}(e^{j\omega}) = \frac{1}{2} [X(e^{j\omega}) + X^*(e^{j\omega})]$$

$$X_{\text{ODD}}(e^{j\omega}) = \frac{1}{2} [X(e^{j\omega}) - X^*(e^{j\omega})]$$

$$X(e^{j\omega}) = X_{\text{EVEN}}(e^{j\omega}) + X_{\text{ODD}}(e^{j\omega})$$

$$X_{\text{EVEN}}(e^{j\omega}) = X_{\text{EVEN}}^*(e^{-j\omega})$$

$$X_{\text{ODD}}(e^{j\omega}) = -X_{\text{ODD}}^*(e^{-j\omega})$$

SYMMETRY

FOR REAL DISCRETE SIGNALS

$$x[n] \rightarrow X(e^{j\omega})$$

$$x^*[n] \rightarrow X^*(e^{-j\omega})$$

$$x^*[-n] \rightarrow X^*(e^{j\omega})$$

$$\text{Re}\{x[n]\} \rightarrow X_{\text{even}}(e^{j\omega})$$

$$j \text{Im}\{x[n]\} \rightarrow X_{\text{odd}}(e^{j\omega})$$

$$x_{\text{even}}[n] \rightarrow \text{Re}\{X(e^{j\omega})\} = X_{\text{Re}}(e^{j\omega})$$

$$x_{\text{odd}}[n] \rightarrow j \text{Im}\{X(e^{j\omega})\} = jX_{\text{Im}}(e^{j\omega})$$

$$x[n] \rightarrow X(e^{j\omega})$$

$$X(e^{j\omega}) = X^*(e^{-j\omega}) \quad \text{Conjugate symmetric}$$

$$X_{\text{Re}}(e^{j\omega}) = X_{\text{Re}}(e^{-j\omega}) \quad \text{Real part is even}$$

$$X_{\text{Im}}(e^{j\omega}) = -X_{\text{Im}}(e^{-j\omega}) \quad \text{Imaginary part is odd}$$

$$|X(e^{j\omega})| = |X(e^{-j\omega})| \quad \text{Magnitude is even}$$

$$\angle X(e^{-j\omega}) = -\angle X(e^{j\omega}) \quad \text{Phase is odd}$$

$$x_{\text{even}}[n] \rightarrow X_{\text{Re}}(e^{j\omega})$$

$$x_{\text{odd}}[n] \rightarrow jX_{\text{Im}}(e^{j\omega})$$

$$x[n] \longrightarrow X(e^{j\omega}) \Rightarrow x^*[n] \longrightarrow X^*(e^{-j\omega})$$

PROOF:

$$X^*(e^{-j\omega}) = \left(\sum_{n=-\infty}^{+\infty} x[n] e^{j\omega n} \right)^* = \sum_{n=-\infty}^{+\infty} x^*[n] e^{-j\omega n} = Y(e^{j\omega})$$

$\uparrow$
 $Y[n] := x^*[n]$

Ex

DTFT of the **real** sequence $x[n] = a^n u[n]$

$$X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}} = X^*(e^{-j\omega})$$

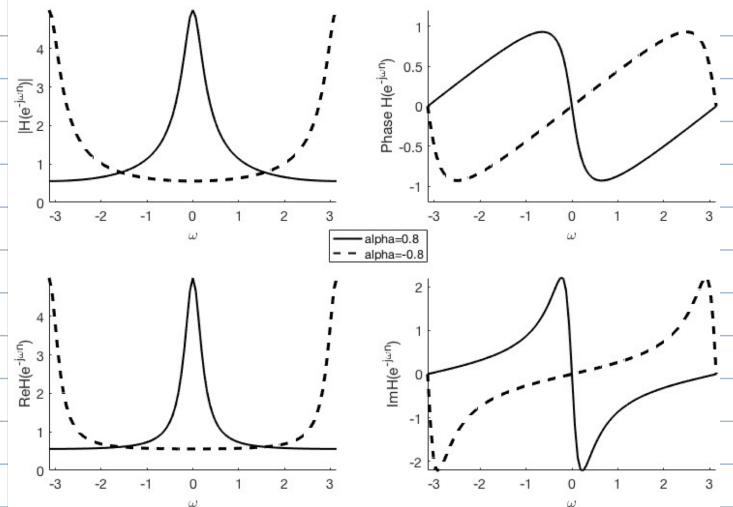
$$|a| < 1$$

$$X_{\text{Re}}(e^{j\omega}) = \frac{1 - a \cos(\omega)}{1 + a^2 - 2a \cos(\omega)} = X_{\text{Re}}(e^{-j\omega})$$

$$X_{\text{Im}}(e^{j\omega}) = \frac{-a \sin(\omega)}{1 + a^2 - 2a \cos(\omega)} = -X_{\text{Im}}(e^{-j\omega})$$

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1 + a^2 - 2a \cos(\omega)}} = |X(e^{-j\omega})|$$

$$\angle X(e^{j\omega}) = \tan^{-1} \left(\frac{-a \sin(\omega)}{1 - a \cos(\omega)} \right) = -\angle X(e^{-j\omega})$$



TIME SHIFT

$$x[n] \longrightarrow X(e^{j\omega}) \Leftrightarrow x[n - n_0] \longrightarrow X(e^{j\omega}) e^{-j\omega n_0}$$

PROOF:

$$X_s(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n - n_0] e^{-j\omega n} = \sum_{m=-\infty}^{+\infty} x[m] e^{-j\omega m} e^{-j\omega n_0} = X(e^{j\omega}) e^{-j\omega n_0} \quad \#$$

$\uparrow$
 $m = n - n_0$
 $n = m + n_0$

FREQUENCY SHIFT

$$x[n] \leftrightarrow X(e^{j\omega}) \Leftrightarrow x[n] e^{j\omega_0 n} \leftrightarrow X(e^{j(\omega - \omega_0)})$$

PROOF:

$$x[n] e^{j\omega_0 n} \longrightarrow X_s(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} e^{j\omega_0 n} = \sum_{n=-\infty}^{+\infty} x[n] e^{-j(\omega - \omega_0) n} = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = X(e^{j\omega}) = X(e^{j(\omega - \omega_0)}) \quad \#$$

$\uparrow$
 $\omega := \omega - \omega_0$

TIME REVERSAL

$$x[n] \leftrightarrow X(e^{j\omega}) \Leftrightarrow x[-n] \leftrightarrow X(e^{-j\omega})$$

Proof:

$$X_p(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[-n] e^{-j\omega(-n)} = \sum_{\substack{\gamma=-\infty \\ \gamma:=-n}}^{+\infty} x[\gamma] e^{-j\omega\gamma} = X(e^{-j\omega}) \quad \#$$

FREQUENCY DERIVATION

$$x[n] \longleftrightarrow X(e^{j\omega}) \iff n x[n] \longleftrightarrow j \frac{d}{d\omega} X(e^{j\omega})$$

Proof:

$$\frac{d}{d\omega} X(e^{j\omega}) = \frac{d}{d\omega} \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} x[n] (-jn) e^{-j\omega n} = -j \sum_{n=-\infty}^{+\infty} n x[n] e^{-j\omega n} = -j \sum_{n=-\infty}^{+\infty} \gamma[n] e^{-j\omega n} \quad \#$$

$\gamma[n] = n x[n]$ } SOMEHOW

CONVOLUTION

$$\left. \begin{array}{l} x[n] \longleftrightarrow X(e^{j\omega}) \\ h[n] \longleftrightarrow H(e^{j\omega}) \end{array} \right\} \iff y[n] = x[n] * h[n] \iff Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$$

Proof:

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} y[n] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} x[k] h[n-k] e^{-j\omega n} = \sum_{k=-\infty}^{+\infty} x[k] \underbrace{\sum_{n=-\infty}^{+\infty} h[n-k] e^{-j\omega n}}_{H(e^{j\omega}) e^{-j\omega k}} = \\ &= H(e^{j\omega}) \sum_{k=-\infty}^{+\infty} x[k] e^{-j\omega k} = H(e^{j\omega}) X(e^{j\omega}) \quad \# \end{aligned}$$

WINDOW 6

$$\left. \begin{array}{l} x[n] \longleftrightarrow X(e^{j\omega}) \\ h[n] \longleftrightarrow H(e^{j\omega}) \end{array} \right\} \iff y[n] = x[n] h[n] \iff Y(e^{j\omega}) = X(e^{j\omega}) *_{\omega} H(e^{j\omega})$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) H(e^{j(\omega-\theta)}) d\theta$$

PARSEVAL'S THEOREM

PARSEVAL'S THEOREM STATES THAT THE SUM OF THE PRODUCTS OF INDIVIDUAL SAMPLES IN TIME IS THE SAME AS THAT IN FREQUENCY (APART FROM A COEFFICIENT).

$$\sum_{n=-\infty}^{+\infty} x[n] y^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) Y^*(e^{j\omega}) d\omega$$

PARSEVAL'S THEOREM TELLS US THAT THE DFT IS A LINEAR TRANSFORM THAT PRESERVES THE NORM OF A SIGNAL (UP TO A FACTOR OF $1/\sqrt{2\pi}$).

THEREFORE WE CAN THINK OF THE DFT AS A ROTATION IN THE INFINITELY-MANY-DIMENSIONAL SPACE.

THIS CAN BE USED TO COMPUTE THE ENERGY OF A SIGNAL

$$\sum_{n=-\infty}^{+\infty} x[n] x^*[n] = \sum_{n=-\infty}^{+\infty} |x[n]|^2 = E_x = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(e^{j\omega}) x^*(e^{j\omega}) d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x(e^{j\omega})|^2 d\omega$$

Ex

$$x[n] = \delta[n]$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} \delta[n] e^{-j\omega n} = \sum_{n=0}^0 1 \cdot e^{-j\omega 0} = 1$$

SPECTRUM OF A DELTA IS CONSTANT FOR ALL FREQUENCIES
"WHITE NOISE"

$$x[n] = 3\delta[n+2] + \delta[n] + 2\delta[n-1] - \delta[n-3]$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = 3e^{j2\omega} + 1 + 2e^{-j\omega} - e^{-j3\omega}$$

DELAYS IN TIME RESULT IN PHASE SHIFTS IN FREQUENCY

FREQUENCY RESPONSE

THE DFT OF THE IMPULSE RESPONSE OF AN LTI SYSTEM IS THE FREQUENCY RESPONSE.

$$\sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n} = H(e^{j\omega})$$

WHERE:

$|H(e^{j\omega})|$ AMPLITUDE RESPONSE

$\angle H(e^{j\omega})$ PHASE RESPONSE

Ex

FIND THE FREQUENCY RESPONSE (1st of these col)

$$y[n] = x[n-n_0] \rightarrow h[n] = \delta[n-n_0]$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n} = e^{-j\omega n_0}$$

$|H(e^{j\omega})| = 1$
 $\angle H(e^{j\omega}) = -\omega n_0$

NO CHANGE IN AMPLITUDE

PHASE SHIFT

$$y[n] = \sum_{k=0}^{N-1} A x[n-k] \rightarrow h[n] = \sum_{k=0}^{N-1} A \delta[n-k] = A(u[n] - u[n-N])$$

$$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$$

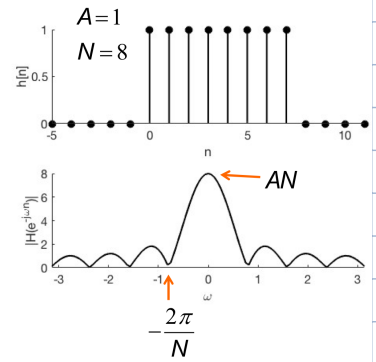
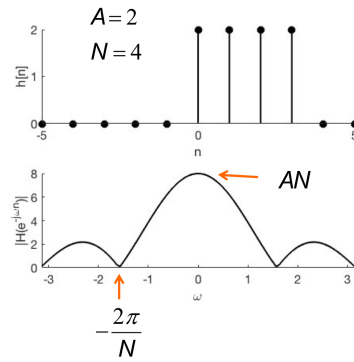
$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} \left(\sum_{k=0}^{N-1} A \delta[n-k] \right) e^{-j\omega n} = \sum_{n=0}^{N-1} A e^{-j\omega n} = A \frac{1-e^{-j\omega N}}{1-e^{-j\omega}}$$

$$A \frac{e^{-j\omega \frac{N}{2}}}{e^{-j\omega \frac{N}{2}}} \frac{e^{j\omega \frac{N}{2}} - e^{-j\omega \frac{N}{2}}}{e^{j\omega \frac{N}{2}} - e^{-j\omega \frac{N}{2}}} = A \frac{e^{-j\omega \frac{N}{2}}}{e^{-j\omega \frac{N}{2}}} \frac{2j \sin(\omega \frac{N}{2})}{2j \sin(\omega \frac{1}{2})} = A e^{-j\omega (\frac{N}{2} - \frac{1}{2})} \frac{\sin(\omega \frac{N}{2})}{\sin(\omega \frac{1}{2})}$$

(sin(N/2)) (sin(1/2))

$$|H(e^{j\omega})| = \begin{cases} |A|N & \omega=0 \\ |A| \left| \frac{\sin(\frac{\omega N}{2})}{\sin(\frac{\omega}{2})} \right| & \text{OTHERWISE} \end{cases}$$

$$\angle H(e^{j\omega}) = \angle A - \frac{\omega}{2}(N-1) + \left[\frac{\sin(\frac{\omega N}{2})}{\sin(\frac{\omega}{2})} \right]$$



ZEROS IN $\omega = \frac{2k\pi}{N}$

$$\sin\left(\frac{k\pi}{N}\right) = 0 \rightarrow \frac{\omega N}{2} = k\pi \rightarrow \omega = \frac{2k\pi}{N}$$

EX

$$y[n] = \sum_{k=0}^{+\infty} a^k x[n-k] \rightarrow h[n] = \sum_{k=0}^{+\infty} a^k \delta[n-k] = a^n u[n]$$

$$|H(e^{j\omega})| = \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n} = \sum_{n=0}^{+\infty} a^n e^{-j\omega n} = \sum_{n=0}^{+\infty} (a e^{-j\omega})^n = \frac{1}{1 - a e^{-j\omega}}, \quad |a| < 1$$

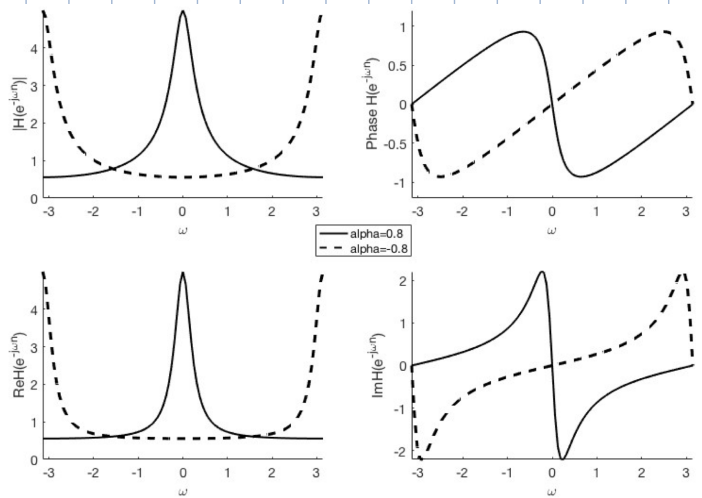
SUPPOSING $a \in \mathbb{R}$:

$$X_{\text{Re}}(e^{j\omega}) = \frac{1 - a \cos(\omega)}{1 + a^2 - 2a \cos(\omega)}$$

$$X_{\text{Im}}(e^{j\omega}) = \frac{-a \sin(\omega)}{1 + a^2 - 2a \cos(\omega)}$$

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1 + a^2 - 2a \cos(\omega)}}$$

$$\angle X(e^{j\omega}) = \tan^{-1} \left(\frac{-a \sin(\omega)}{1 - a \cos(\omega)} \right)$$



CONVERGENCE

SINCE THE DTFT IS AN INFINITE SUM, IT DOES NOT ALWAYS CONVERGE. WE NEED THEREFORE TO GUARANTEE CONVERGENCE FOR IT TO EXIST.

THERE ARE MANY KNOWN SUFFICIENT CONDITIONS FOR THE DTFT TO CONVERGE.

ABSOLUTE SUMMABILITY

ABSOLUTE SUMMABILITY OF $x[n]$ IS A SUFFICIENT CONDITION FOR THE EXISTENCE OF THE DTFT.

$$|X(e^{j\omega})| < \infty \Leftrightarrow |X(e^{j\omega})| = \left| \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{+\infty} |x[n]| |e^{-j\omega n}| = \sum_{n=-\infty}^{+\infty} |x[n]| < \infty$$

NOTE THAT THIS CONDITION IS SUFFICIENT BUT NOT NECESSARY. THE DFT $X(e^{j\omega})$ OF $x[n]$ EXISTS IF IT CONVERGES IN SOME SENSE (WHICH MIGHT BE DIFFERENT FROM THIS ONE).

UNIFORM CONVERGENCE

ONE CONDITION, CALLED "UNIFORM CONVERGENCE CONDITION", IS THE FOLLOWING:

LET $X_M(e^{j\omega})$ DENOTE THE PARTIAL SUM OF WEIGHTED COMPLEX EXPONENTIALS.

$$X_M(e^{j\omega}) = \sum_{n=-M}^M x[n] e^{-j\omega n}$$

THEN, CONVERGENCE IS GUARANTEED IF

$$\lim_{M \rightarrow \infty} |X(e^{j\omega}) - X_M(e^{j\omega})| = 0$$

THIS CONDITION IS NOT VALID FOR SEQUENCES THAT ARE NOT ABSOLUTELY SUMMABLE. FOR THIS REASON WE INTRODUCE THE "MEAN SQUARE CONVERGENCE" CONDITION.

MEAN SQUARE CONVERGENCE

ABSOLUTELY SUMMABLE SEQUENCES ALWAYS HAVE FINITE ENERGY, HOWEVER THE OPPOSITE IS NOT TRUE, FINITE ENERGY SEQUENCES ARE NOT NECESSARILY ABSOLUTELY SUMMABLE.

$$\text{ABS. SUMMABILITY} \xRightarrow{\text{}} \text{FINITE ENERGY}$$

SOME FINITE ENERGY (SQUARE SUMMABLE) SEQUENCES ACHIEVE MEAN SQUARE CONVERGENCE:

$$\lim_{M \rightarrow \infty} \int_{-\pi}^{\pi} |X(e^{j\omega}) - X_M(e^{j\omega})|^2 d\omega = 0$$

NOTE THAT IT IS THE TOTAL ENERGY OF THE ERROR THAT MUST APPROACH ZERO, NOT THE ERROR ITSELF.

SPECIAL CASES

THE FOURIER TRANSFORM CAN ALSO BE DEFINED FOR A CERTAIN CLASS OF SEQUENCES THAT ARE NEITHER ABSOLUTELY SUMMABLE NOR SQUARE SUMMABLE.

SOME OF THESE SEQUENCES ARE:

- HEAVISIDE STEP SEQUENCE
- SINUSOIDAL SEQUENCE
- COMPLEX EXPONENTIAL SEQUENCE

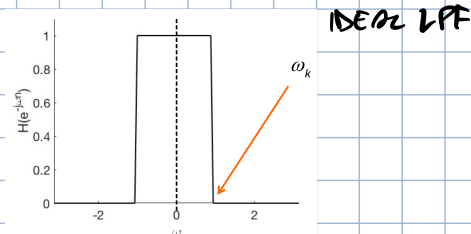
THESE SEQUENCES ARE REPRESENTABLE VIA THE DFT USING THE DIRAC DELTA

$$\delta(t) = \begin{cases} \infty & t=0 \\ 0 & \text{OTHERWISE} \end{cases} \quad \cup \quad \int_{-\infty}^{+\infty} \delta(t) dt = 1$$

Sequence	DTFT
$\delta[n-n_0]$	$e^{-j\omega n_0}$
1	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad a < 1$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$
$\cos(\omega_0 n + \phi)$	$\sum_{k=-\infty}^{\infty} [\pi e^{j\phi}\delta(\omega - \omega_0 + 2\pi k) + \pi e^{-j\phi}\delta(\omega + \omega_0 + 2\pi k)]$

INVERSE APPROXIMATIONS

$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \omega_k \\ 0 & \text{otherwise} \end{cases}, \quad \omega_k \in [-\pi, \pi]$$



$$h[n] = ?$$

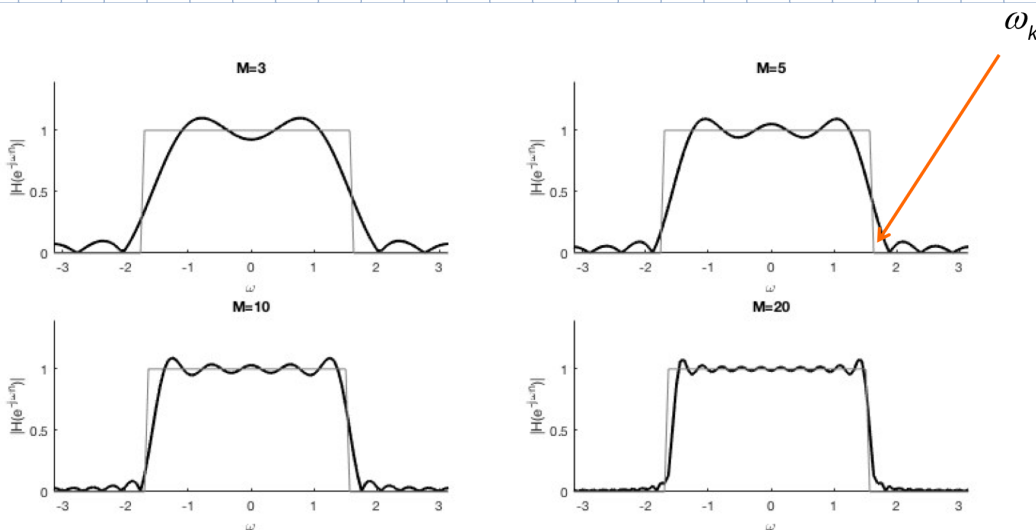
$$h[n] = \frac{1}{2\pi} \int_{-\omega_k}^{\omega_k} H(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_k}^{\omega_k} e^{j\omega n} d\omega = \frac{1}{2\pi} \left. \frac{e^{j\omega n}}{jn} \right|_{-\omega_k}^{\omega_k} = \frac{e^{j\omega_k n} - e^{-j\omega_k n}}{2\pi j n} = \frac{\sin(\omega_k n)}{\pi n}$$

$\frac{e^{jx} - e^{-jx}}{2j} = \sin(x)$

$$h[n] = \frac{\sin(\omega_k n)}{\pi n} \longrightarrow H'(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \frac{\sin(\omega_k n)}{\pi n} e^{-j\omega n}$$

BUT $H'(e^{j\omega})$ IS AN INFINITE SUM OF SIGNALS. LET'S SEE WHAT HAPPENS IF WE APPROXIMATE IT WITH A FINITE NUMBER OF SUMS:

$$H_M(e^{j\omega}) = \sum_{n=-M}^M \frac{\sin(\omega_k n)}{\pi n} e^{-j\omega n}$$



NOTES:

- TRANSITIONS GET SHARPER WITH HIGHER M , SINCE WE ARE ADDING HIGH FREQUENCY COMPONENTS

- APPROXIMATION WITH RIPPLES IN THE CONSTANT PASS-BAND REGARDLESS OF M .

THE NUMBER OF RIPPLES INCREASES WITH M AND THE HEIGHT OF THE LARGEST RIPPLES

REMAINS THE SAME FOR ALL VALUES OF M .

THIS OSCILLATORY BEHAVIOR IS COMMONLY KNOWN AS "GIBBS PHENOMENON"

- CONVERGENCE WILL BE REACHED IN THE SENSE OF MEAN SQUARE

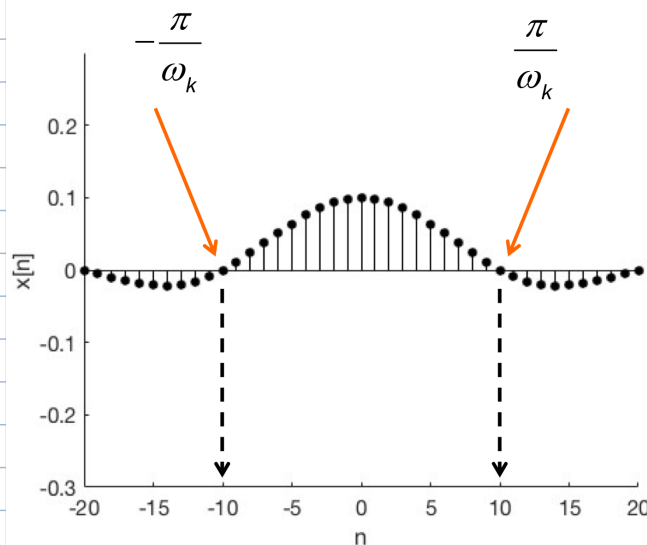
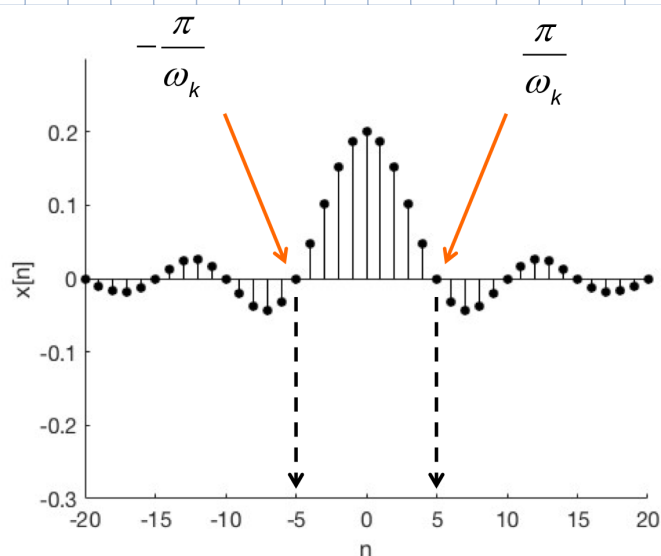
NOTE THAT $H'(e^{j\omega})$ HAS A PROBLEM IN $n=0$

$$H'(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} \frac{\sin(\omega_k n)}{\pi n} e^{-j\omega n}$$

TO SOLVE IT WE CAN COMPUTE THE IDFT IN ω USING THE DEFINITION

$$h[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega \cdot 0} d\omega = \frac{1}{2\pi} \int_{-\omega_k}^{\omega_k} d\omega = \frac{\omega_k - (-\omega_k)}{2\pi} = \frac{\omega_k}{\pi} \neq 0 \rightarrow \text{OK}$$

$$h[n] = \begin{cases} \frac{\sin(\omega_k n)}{\pi n} & n \neq 0 \\ \frac{\omega_k}{\pi} & n = 0 \end{cases} = \frac{\omega_k}{\pi} \frac{\sin(\omega_k n)}{\pi n} \stackrel{\omega_k = 2\pi f_k}{=} 2f_k \text{sinc}(2f_k n)$$



FOR A LARGER BANDWIDTH IN THE FREQUENCY DOMAIN ω_k MUST BE LARGER AND THEREFORE THE ZEROS SHRINK IN THE TIME DOMAIN

$\hookrightarrow$ ~ CAN THINK OF AS IT MUST BE REACTIVE TO HIGHER FREQUENCY SIGNALS / MORE VERTICAL STUFF